**ON HMS *BEAGLE* DARWIN** FILLED **15 NOTEBOOKS** WITH SKETCHES AND OBSERVATIONS

***ON THE ORIGIN OF SPECIES*** **SOLD 1,250 COPIES** ON THE DAY OF PUBLICATION

mockingbirds, finches, and giant tortoises, for example—shared very similar characteristics, he began to think that in each case they may have shared a common ancestor.

On his return to the UK, Darwin continued to develop his ideas. He was strongly influenced by *An Essay on the Principle of Population* by Thomas Malthus, which predicted that human overpopulation would lead to a struggle for survival due to limited food resources. Darwin wondered if this would apply to animals, too.

**Competition and evolution**

By 1838, Darwin had formulated his theory of evolution, but he knew it would cause outrage, as it contradicted the Christian view of creation. He was cautious about publishing and instead spent the next 20 years gathering more supporting evidence for his theory. Then, in 1858, he received a manuscript from Alfred Russel Wallace (see box), who had come up with a similar idea. A year later, Darwin published *On the Origin of Species*. The book sold out on the first day of publication.

Darwin's theory of evolution by natural selection was based on his realization that there is always some variation between all the individuals in a species and that more individuals are born than survive. He suggested that, in the competition for survival, it is the characteristics of an individual that make the difference and that only the fittest survive. For example, those with thicker fur are far more likely to survive in a cold climate and go on to produce offspring. If a useful trait, such as thick fur, can be inherited, more of the next generation will possess that trait. Over time, such small changes add up to a large and noticeable difference.

Although Darwin made sure not to discredit Christian beliefs in his book, it was still a profound shock to Victorian society. His work commanded huge respect nonetheless. When he died, Darwin was given a state funeral and was buried in Westminster Abbey.

***Darwin noticed** that the shapes of the beaks of the Galápagos finches varied in accordance with their diets. He proposed that the finches were linked by one common ancestor from which new species evolved as they adapted to the conditions and food of their inhabited islands.*